



Qualification Pack

Two/Three-Wheeler Mechanic (Including LPG & CNG)

QP Code: WBSC/ASC/Q1301

Version: 1.0

NSQF Level: 3

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Qualification Pack

Contents

WBSC/ASC/Q1301: Two/Three-Wheeler Mechanic (Including LPG & CNG)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
WBSC/ASC/N1301: Apply Safe Working Practices (Two/Three-Wheeler Mechanic (Including LPG & CNG)).	5
WBSC/ASC/N1302: Identify different types of Two/Three wheeler and their configuration.	9
WBSC/ASC/N1303: Identify and explain the key features of two and three-wheeler engines.	13
WBSC/ASC/N1304: Identify hand tools, measuring tools and their uses.	17
WBSC/ASC/N1305: Troubleshoot any fault of engine system, fuel feed system and transmission system.	21
WBSC/ASC/N1306: Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.	25
WBSC/ASC/N1307: Repair parts like fork, suspension system, steering column, mud guard, lubrication system.	29
WBSC/ASC/N1308: Work in real job situation with special emphasis on basic safety and hazards in this domain (Two/Three-Wheeler Mechanic (Including LPG & CNG)).	32
DGT/VSQ/N0101: Employability Skills (30 Hours)	35
Assessment Guidelines and Weightage	40
<i>Assessment Guidelines</i>	40
<i>Assessment Weightage</i>	41
Acronyms	43
Glossary	44



Qualification Pack

WBSC/ASC/Q1301: Two/Three-Wheeler Mechanic (Including LPG & CNG)

Brief Job Description

The Two/Three-Wheeler Mechanic (Including LPG & CNG) will be responsible for diagnosing, troubleshooting and repairing a variety of two and three-wheeler vehicles, including those with LPG and CNG fuel systems. The role involves working with engines, fuel systems, transmission systems, brakes, suspension, steering and electrical components. Adherence to safety protocols and quality standards is essential.

Personal Attributes

A Two/Three-Wheeler Mechanic (Including LPG & CNG) should be technically skilled, detail-oriented and safety conscious with strong problem-solving abilities. They must be patient and methodical when troubleshooting complex systems and remain calm under pressure in a busy workshop environment. Good hand-eye coordination, physical stamina and the ability to follow technical manuals are essential along with a responsible attitude toward safety procedures and quality standards.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [WBSC/ASC/N1301: Apply Safe Working Practices \(Two/Three-Wheeler Mechanic \(Including LPG & CNG\)\)](#).
2. [WBSC/ASC/N1302: Identify different types of Two/Three wheeler and their configuration.](#)
3. [WBSC/ASC/N1303: Identify and explain the key features of two and three-wheeler engines.](#)
4. [WBSC/ASC/N1304: Identify hand tools, measuring tools and their uses.](#)
5. [WBSC/ASC/N1305: Troubleshoot any fault of engine system, fuel feed system and transmission system.](#)
6. [WBSC/ASC/N1306: Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.](#)
7. [WBSC/ASC/N1307: Repair parts like fork, suspension system, steering column, mud guard, lubrication system.](#)
8. [WBSC/ASC/N1308: Work in real job situation with special emphasis on basic safety and hazards in this domain \(Two/Three-Wheeler Mechanic \(Including LPG & CNG\)\)](#).
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)



Qualification Pack

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automobile Vehicle Service
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
Country	India
NSQF Level	3
Credits	12
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3115.0601,7231.0500,7231.0501
Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass with 2 Years of experience OR Previous relevant Qualification of NSQF Level 2.5 with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	17/12/2027
NSQC Approval Date	17/12/2024
Version	1.0
Reference code on NQR	QG-03-AU-03426-2024-V1-WBSC
NQR Version	1



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WBSC/ASC/N1301: Apply Safe Working Practices (Two/Three-Wheeler Mechanic (Including LPG & CNG)).

Description

Apply Safe Working Practices.

Scope

The scope covers the following :

- Safe Working Practices.
- Unsafe situations according to site policy.
- Fire and safety and fire hazards.
- Different fire extinguishers and use them as per requirements.
- Bubble test for LPG leakage in 3-wheeler.
- CNG leakage by sniffing in Two - Three wheeler.
- Basic pollution control measure used in automobile shop surroundings.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply and maintain Safe Working Practices.
- PC2.** Recognize any unsafe situations according to site policy.
- PC3.** Identify fire and safety and fire hazards.
- PC4.** Identify different fire extinguishers and use them as per requirements.
- PC5.** Perform bubble test for LPG leakage in 3-wheeler.
- PC6.** Identify CNG leakage by sniffing in Two - Three wheeler.
- PC7.** Follow basic pollution control measure used in automobile shop surroundings.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Procedures to achieve a safe working environment in line with occupational health and safety regulations and requirements according to site policy.
- KU2.** Any unsafe situations according to site policy, and assess his report accordingly.
- KU3.** Precautions on fire and safety hazards and report according to site policy and procedures.
- KU4.** Difference between BS & Euro norms.
- KU5.** Basic knowledge about Road Transport & Traffic rules.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Recognize any unsafe situations according to site policy, and assess his report accordingly.
- GS2.** Demonstrate Personal Productive Equipment (PPE) like: safety helmet, safety glove, safety shoe, use the same as per related working environment.
- GS3.** Demonstrate basic first aid & CPR and use them under different circumstances.
- GS4.** Identify different fire extinguishers and use the same as per requirement in a mock drill.
- GS5.** Perform bubble test for LPG leakage in two three wheeler.
- GS6.** Identify CNG leakage by sniffing in two three wheeler.
- GS7.** Identify the differences of BS I, II, III, IV, VI.
- GS8.** Take preventive maintenance on selective parts.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	70	-	-
PC1. Apply and maintain Safe Working Practices.	-	-	-	-
PC2. Recognize any unsafe situations according to site policy.	-	-	-	-
PC3. Identify fire and safety and fire hazards.	-	-	-	-
PC4. Identify different fire extinguishers and use them as per requirements.	-	-	-	-
PC5. Perform bubble test for LPG leakage in 3-wheeler.	-	-	-	-
PC6. Identify CNG leakage by sniffing in Two - Three wheeler.	-	-	-	-
PC7. Follow basic pollution control measure used in automobile shop surroundings.	-	-	-	-
NOS Total	10	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1301
NOS Name	Apply Safe Working Practices (Two/Three-Wheeler Mechanic (Including LPG & CNG)).
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



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WBSC/ASC/N1302: Identify different types of Two/Three wheeler and their configuration.

Description

Identify different types of Two/Three wheeler and their configuration.

Scope

The scope covers the following :

- The basic differences between two wheelers: types of motorcycles, scooter.
- Engine configurations with various transmission systems.
- The uses of various parts of two/three wheeler.
- Different parts of alternative fuel feed system.
- Different frame and body of two and three wheelers.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** State the basic differences between two wheelers: types of motorcycles, scooter.
- PC2.** Comprehend the engine configurations with various transmission systems
- PC3.** Define the uses of various parts of two/three wheeler.
- PC4.** Identify different parts of alternative fuel feed system.
- PC5.** Identify different frame and body of two and three wheelers.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different parts of two- wheeler.
- KU2.** Differences between motorcycles, scooters, auto rickshaw.
- KU3.** The functions of different parts of three-wheeler.
- KU4.** Functions of main parts of an Internal combustion engine- SI engine, CI engine, LPG and CNG dual fuel engine.
- KU5.** Different body structures of two and three-wheeler (layout only).
- KU6.** Safety and regulatory standards.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify different parts of two-wheeler and three wheeler.
- GS2.** Distinguish the difference between motorcycles, scooters with their main parts.



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- GS3.** Distinguish the difference between loading auto rickshaws: passenger vehicle, delivery vehicle, pickup vehicle, tipper vehicle with their main parts.
- GS4.** Locate the different parts and application in the 2/3 wheelers.
- GS5.** Identify different frame of two and three- wheeler Recognize the Tubular Frame, Engine Based Frame, Twin-Spar Frame, Monocoque Frame.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	80	-	-
PC1. State the basic differences between two wheelers: types of motorcycles, scooter.	-	-	-	-
PC2. Comprehend the engine configurations with various transmission systems	-	-	-	-
PC3. Define the uses of various parts of two/three wheeler.	-	-	-	-
PC4. Identify different parts of alternative fuel feed system.	-	-	-	-
PC5. Identify different frame and body of two and three wheelers.	-	-	-	-
NOS Total	20	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1302
NOS Name	Identify different types of Two/Three wheeler and their configuration.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1303: Identify and explain the key features of two and three-wheeler engines.

Description

Identify and explain the key features of two and three-wheeler engines.

Scope

The scope covers the following :

- Identification of the four stroke and working principle of four stroke engine.
- Differentiation between two stroke and four stroke engines.
- Different engine components like cylinders, pistons, crankshafts and their role in engine functionality.
- The difference between four stroke LPG and four stroke NG and four stroke petrol engines

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the four stroke and narrate the working principle of four stroke engine.
- PC2.** Differentiate two stroke and four stroke engines.
- PC3.** Identify different engine components like cylinders, pistons, crankshafts and their role in engine functionality.
- PC4.** Identify the difference four stroke LPG and four stroke NG and four stroke petrol engines.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Difference between two stroke and four stroke engines.
- KU2.** Working principle of four stroke engine.
- KU3.** Arrangements of Cylinders - single cylinder, Inline (2, 3 ,4 cylinder).
- KU4.** Introduction to valve operating mechanism and cranking system.
- KU5.** Exhaust System of Two/three Wheelers, Exhaust Pipe and Header.
- KU6.** Difference between four stroke LPG. Four stroke CNG and four stroke Petrol Engine.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify the working of two stroke and four stroke engines.
- GS2.** Describe working principle of two stroke and four stroke engines.
- GS3.** Identify the different mechanism of (Valve and cranking system) of an engine.



Qualification Pack

- GS4.** Identify the cylinders arrangement of various engine types.
- GS5.** Identify the different components of exhaust system of two/three wheelers.
- GS6.** Recognize the difference between Fuel Delivery System, LPG/ CNG tank, Pressure regulator, gas filters, injectors.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	80	-	-
PC1. Identify the four stroke and narrate the working principle of four stroke engine.	-	-	-	-
PC2. Differentiate two stroke and four stroke engines.	-	-	-	-
PC3. Identify different engine components like cylinders, pistons, crankshafts and their role in engine functionality.	-	-	-	-
PC4. Identify the difference four stroke LPG and four stroke NG and four stroke petrol engines.	-	-	-	-
NOS Total	20	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1303
NOS Name	Identify and explain the key features of two and three-wheeler engines.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1304: Identify hand tools, measuring tools and their uses.

Description

Identify hand tools, measuring tools and their uses.

Scope

The scope covers the following :

- Hand tools used for repairing Two /Three wheeler.
- The purpose and use of hand tools.
- The working principles of measuring instruments and special tools used in auto workshop.
- Appropriate tools for lifting a three-wheeler.
- The measuring instruments with working principle of vernier caliper, caliper, micrometer, scale, feeler gauge etc.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Recognize the hand tools used for repairing Two /Three wheeler, like. Hammer mallet, torque wrench, screw driver, file, etc.
- PC2.** Describe the purpose and use of hand tools.
- PC3.** Plan the working principles of measuring instruments and special tools used in auto workshop.
- PC4.** Apply appropriate tools for lifting a three-wheeler.
- PC5.** Recognize the measuring instruments with working principle of vernier caliper, caliper, micrometer, scale, feeler gauge etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Introduction to Hand tools: Overview of the purpose and importance of hand tools in the maintenance and repair of 2/3 wheelers.
- KU2.** The uses of different measuring instruments (Vernier Caliper, Micrometer, Feeler gauge, Multimeter, Tachometer).
- KU3.** The general-purpose hand tools used in Automobile workshop with the principle.
- KU4.** Importance of using right tools for each task to prevent injury and component damage.
- KU5.** Selection of tools as per task.
- KU6.** Overview of Tyre pressure gauge (Analog / digital).
- KU7.** Overview of air compressor.
- KU8.** Procedure for compare with recommended pressure.
- KU9.** Maintenance of tools.

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** Identify different measuring instruments (Vernier Caliper, Micrometer, Feeler gauge, Multimeter, Tachometer).
- GS2.** Apply general purpose hand tools in Automobile workshop. Measurement of length, depth, diameter (inside & outside).
- GS3.** Measure Spark plug gap and valve tappet clearance.
- GS4.** Measure of voltage, currents & resistance.
- GS5.** Measure of RPM.
- GS6.** Test the vehicle electrical system, including battery voltage, testing fuses and diagnosing electrical faults.
- GS7.** Check Tyre pressure and alignment.
- GS8.** Clean and maintain hand and measuring tools.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	80	-	-
PC1. Recognize the hand tools used for repairing Two /Three wheeler, like. Hammer, mallet, torque wrench, screw driver, file, etc.	-	-	-	-
PC2. Describe the purpose and use of hand tools.	-	-	-	-
PC3. Plan the working principles of measuring instruments and special tools used in auto workshop.	-	-	-	-
PC4. Apply appropriate tools for lifting a three-wheeler.	-	-	-	-
PC5. Recognize the measuring instruments with working principle of vernier caliper, caliper, micrometer, scale, feeler gauge etc.	-	-	-	-
NOS Total	20	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1304
NOS Name	Identify hand tools, measuring tools and their uses.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1305: Troubleshoot any fault of engine system, fuel feed system and transmission system.

Description

Troubleshoot any fault of engine system, fuel feed system and transmission system.

Scope

The scope covers the following :

- Basic problems in an engine and their cause.
- Dismantling, replacement of any parts and reassembling of different parts of an engine.
- Basic problems in a carburetor, clutch, gearbox and their cause.
- Dismantling, replacement of any parts and reassembling a clutch, dismantle, replace any parts and reassemble a gearbox.
- Repairing of sprocket system of a motor cycle, adjust primary and secondary chain
- Identification and changing of the components as required viz. Belts, oil and air filters etc.
- Different alternative fuel feed system required in 3 wheeler.
- Replacement of CNG kit/LPG kit in a 3-wheeler and test it for its functionality

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify basic problems in an engine and their cause.
- PC2.** Dismantle, replace any parts and reassemble different parts of an engine.
- PC3.** Identify basic problems in a carburetor and their cause.
- PC4.** Diagnose basic problem of a clutch and their causes.
- PC5.** Diagnose basic problem of a gearbox and their causes.
- PC6.** Dismantle, replace any parts and reassemble a clutch, dismantle, replace any parts and reassemble a gearbox.
- PC7.** Repair sprocket system of a motor cycle, adjust primary and secondary chain.
- PC8.** Identify and change the components as required viz. Belts, oil and air filters etc.
- PC9.** Demonstrate different alternative fuel feed system required in 3wheeler.
- PC10.** Replace CNG kit/LPG kit in a 3-wheeler and test it for its functionality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The basic problems in an engine and find the cause.
- KU2.** Dismantle any parts and precautions to be taken.
- KU3.** Replacement procedure and reassemble of different parts of an engine.
- KU4.** The basic problems in a carburetor, clutch, gearbox and the cause.



Qualification Pack

- KU5.** Dismantling and replacement procedure for the same parts with precautions.
- KU6.** Procedure to change the components viz. Belts, oil, filters etc.
- KU7.** Different alternative fuel feed system in 3-wheeler.
- KU8.** The function of CNG kit/LPG kit in a three-wheeler.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Recognize basic problems in a fuel feed system and their causes.
- GS2.** Dismantle different parts of a fuel feed system.
- GS3.** Clean, Inspect & refit different parts of a fuel feed system.
- GS4.** Recognize basic problems in alternating fuel feed system (LPG/CNG).
- GS5.** Dismantle, Clean, inspect & refit different parts of Multi valve, LPR, HPR.
- GS6.** Clean, inspect and refit the different parts of multi valve, LPR and HPR.
- GS7.** Dismantle, Clean, inspect & refit of LPG and CNG kit.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	130	-	-
PC1. Identify basic problems in an engine and their cause.	-	-	-	-
PC2. Dismantle, replace any parts and reassemble different parts of an engine.	-	-	-	-
PC3. Identify basic problems in a carburetor and their cause.	-	-	-	-
PC4. Diagnose basic problem of a clutch and their causes.	-	-	-	-
PC5. Diagnose basic problem of a gearbox and their causes.	-	-	-	-
PC6. Dismantle, replace any parts and reassemble a clutch, dismantle, replace any parts and reassemble a gearbox.	-	-	-	-
PC7. Repair sprocket system of a motor cycle, adjust primary and secondary chain.	-	-	-	-
PC8. Identify and change the components as required viz. Belts, oil and air filters etc.	-	-	-	-
PC9. Demonstrate different alternative fuel feed system required in 3wheeler.	-	-	-	-
PC10. Replace CNG kit/LPG kit in a 3-wheeler and test it for its functionality.	-	-	-	-
NOS Total	30	130	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1305
NOS Name	Troubleshoot any fault of engine system, fuel feed system and transmission system.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1306: Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.

Description

Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.

Scope

The scope covers the following :

- Basic problem of brake system and their causes.
- Dismantling, replacement of any parts and reassembling of a brake shoe/disc brake.
- Basic problems of wheel system and their causes.
- Dismantling, replacement of any parts and reassembling of a wheel.
- Basic problems of ignition system and their causes.
- Dismantling, replacement of any parts and reassembling of ignition system.
- Basic problems of battery and their causes.
- Replacement/ repairing of the connection of battery.
- Basic problems of lighting system, switch, relay, and their causes.
- Repairing of lighting system, switches, relays.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Diagnose basic problem of brake system and their causes.
- PC2.** Dismantle, replace any parts and reassemble a brake shoe/disc brake.
- PC3.** Diagnose basic problems of wheel system and their causes.
- PC4.** Dismantle, replace any parts and reassemble a wheel.
- PC5.** Diagnose basic problems of ignition system and their causes.
- PC6.** Dismantle, replace any parts and reassemble ignition system.
- PC7.** Diagnose basic problems of battery and their causes.
- PC8.** Replace/ repair the connection of battery.
- PC9.** Diagnose basic problems of lighting system, switch, relay, and their causes.
- PC10.** Repair lighting system, switches, relays.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic problem of brake system and their causes.
- KU2.** Procedure of dismantling, Cleaning, Inspecting & assembling Brake assembly.
- KU3.** Trouble shooting of Brake system.
- KU4.** Brake bleeding process.



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- KU5.** Basic problems of ignition system and their causes.
- KU6.** Dismantling, Cleaning, replacing parts of ignition system.
- KU7.** Basic problems of battery and their causes.
- KU8.** Describe the procedure of replacing a Battery in a correct sequence.
- KU9.** Basic problems of lighting system, switch, relay, and their causes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Recognize basic problem of brake system and their causes.
- GS2.** Dismantle, Clean, Inspect & assemble Brake assembly.
- GS3.** Trouble shoot Brake system.
- GS4.** Demonstrate Brake bleeding process.
- GS5.** Check Toe in & Toe out in 3 wheeler.
- GS6.** Overhaul suspension system.
- GS7.** Recognize basic problems of ignition system and their causes.
- GS8.** Dismantle, Clean, replace parts of ignition system.
- GS9.** Identify basic problems of battery and their causes.
- GS10.** Replace a Battery in a correct sequence.
- GS11.** Identify basic problems of lighting system, switch, relay, and their causes.
- GS12.** Trouble shoot Starter motor.
- GS13.** Trouble shoot electrical accessories.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	130	-	-
PC1. Diagnose basic problem of brake system and their causes.	-	-	-	-
PC2. Dismantle, replace any parts and reassemble a brake shoe/disc brake.	-	-	-	-
PC3. Diagnose basic problems of wheel system and their causes.	-	-	-	-
PC4. Dismantle, replace any parts and reassemble a wheel.	-	-	-	-
PC5. Diagnose basic problems of ignition system and their causes.	-	-	-	-
PC6. Dismantle, replace any parts and reassemble ignition system.	-	-	-	-
PC7. Diagnose basic problems of battery and their causes.	-	-	-	-
PC8. Replace/ repair the connection of battery.	-	-	-	-
PC9. Diagnose basic problems of lighting system, switch, relay, and their causes.	-	-	-	-
PC10. Repair lighting system, switches, relays.	-	-	-	-
NOS Total	30	130	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1306
NOS Name	Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1307: Repair parts like fork, suspension system, steering column, mud guard, lubrication system.

Description

Repair parts like fork, suspension system, steering column, mud guard, lubrication system.

Scope

The scope covers the following :

- Basic problems of fork, suspension system, steering column, mud guard, lubrication system and their causes.
- Replacement/ repairing of the fork, suspension system, steering column, mud guard, lubrication system.
- Restoration of the suspension systems efficiency.
- Proper lubrication of moving parts, preventing wear and tear.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Diagnose basic problems of fork, suspension system, steering column, mud guard, lubrication system and their causes.
- PC2.** Replace/ repair the fork, suspension system, steering column, mud guard, lubrication system.
- PC3.** Restore the suspension systems efficiency.
- PC4.** Apply proper lubrication of moving parts, preventing wear and tear.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The procedure to find the basic problems of fork, suspension system, steering column, mud guard.
- KU2.** The system of lubrication and replacement of lubrication in timely manner.
- KU3.** The procedure for repairing the fork, suspension system, steering column and mud guard.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Diagnose basic problems of fork.
- GS2.** Diagnose suspension system, steering column, mud guard.
- GS3.** Check the lubrication system and replace the lubrication in timely manner.
- GS4.** Replace/ repair the fork, suspension system, steering column, mud guard, lubrication system.
- GS5.** Diagnose and troubleshoot of handle bar steering system.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	80	-	-
PC1. Diagnose basic problems of fork, suspension system, steering column, mud guard, lubrication system and their causes.	-	-	-	-
PC2. Replace/ repair the fork, suspension system, steering column, mud guard, lubrication system.	-	-	-	-
PC3. Restore the suspension systems efficiency.	-	-	-	-
PC4. Apply proper lubrication of moving parts, preventing wear and tear.	-	-	-	-
NOS Total	20	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1307
NOS Name	Repair parts like fork, suspension system, steering column, mud guard, lubrication system.
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

WBSC/ASC/N1308: Work in real job situation with special emphasis on basic safety and hazards in this domain (Two/Three-Wheeler Mechanic (Including LPG & CNG)).

Description

Work in real job situation with special emphasis on basic safety and hazards in this domain.

Scope

The scope covers the following :

- Work in the real job situation.
- Report preparation on the competency developed.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- PC2.** Prepare report for this component of training of the course to ensure whether competency has been developed.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- GS2.** Prepare report for this component of training of the course to ensure whether competency has been developed.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	150	-	-
PC1. Work in the real job situation with special emphasis on basic safety and hazards in this domain.	-	-	-	-
PC2. Prepare report for this component of training of the course to ensure whether competency has been developed.	-	-	-	-
NOS Total	-	150	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/ASC/N1308
NOS Name	Work in real job situation with special emphasis on basic safety and hazards in this domain (Two/Three-Wheeler Mechanic (Including LPG & CNG)).
Sector	Automotive
Sub-Sector	
Occupation	Two/Three-Wheeler Mechanic (Including LPG & CNG)
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment will be based on the concept of Independent Assessors empaneled with West Bengal State Council of Technical & Vocational Education & Skill Development (WBSCT&VE&SD), identified, selected, trained and certified on Assessment techniques. These Assessors would be aligned to assess as per the laid down criteria.

WBSCT&VE&SD would conduct assessment only at the training centers or designated testing centers authorized by WBSCT&VE&SD. Ideally, the assessment will be a continuous process comprising of two distinct steps:

- Continuous assessment by Trainers
- Term end /Final Assessment by WBSCT&VE&SD

Each National Occupational Standard (NOS) in the respective QPs will be assigned weightage. Each Performance Criteria in the NOS will be assigned marks for theory and/or practical based on relative importance and criticality of function. This will facilitate preparation of question bank / paper sets for each of the QPs. Each of these papers sets/question banks created by subject matter experts through WBSCT&VE&SD, especially with regard to the practical test and the defined tolerances, finish, accuracy



Qualification Pack

etc. The following tools are proposed to be used for final assessment:

i. Written Test: This will comprise of (i) True/False Statements and/or (ii) Multiple Choice Questions and/or (iii) Matching Type Questions. Online system for this will be preferred.

ii. Practical Test: This will comprise a test job to be prepared as per project briefing following appropriate working steps, using necessary tools, equipment and instruments. Through observation it will be possible to ascertain candidate's aptitude, attention to details, quality consciousness etc.

iii. Structured Viva-voce: This tool will be used to assess the conceptual understanding and the behavioral aspects as regards the job role and the specific task at hand.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/ASC/N1301. Apply Safe Working Practices (Two/Three-Wheeler Mechanic (Including LPG & CNG)).	10	70	-	-	80	8
WBSC/ASC/N1302. Identify different types of Two/Three wheeler and their configuration.	20	80	-	-	100	10
WBSC/ASC/N1303. Identify and explain the key features of two and three-wheeler engines.	20	80	-	-	100	10
WBSC/ASC/N1304. Identify hand tools, measuring tools and their uses.	20	80	-	-	100	10
WBSC/ASC/N1305. Troubleshoot any fault of engine system, fuel feed system and transmission system.	30	130	-	-	160	16



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/ASC/N1306.Troubleshoot and repair the fault of brake and wheel, ignition system, electric system.	30	130	-	-	160	16
WBSC/ASC/N1307.Repair parts like fork, suspension system, steering column, mud guard, lubrication system.	20	80	-	-	100	10
WBSC/ASC/N1308.Work in real job situation with special emphasis on basic safety and hazards in this domain (Two/Three-Wheeler Mechanic (Including LPG & CNG)).	-	150	-	-	150	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	830	-	-	1000	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NQR	National Qualification Register



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.



Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
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Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	National Occupational Standard (NOS) are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.



Qualification Pack

Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
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Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf .